

SMART INDIA HACKATHON 2026



TITLE PAGE

- Problem Statement ID – **SIH26209**
- Problem Statement Title- **Student Innovation — Space Technology**
- Theme- **Space Technology**
- PS Category- **Software**
- Team ID- **< fill after portal registration >**
- Team Name (Registered on portal)- **VaanEye**



VaanEye



THE EYE IN THE SKY



IDEA TITLE



- Proposed Solution (Describe your Idea/Solution/Prototype)
- Detailed explanation of the proposed solution
- How it addresses the problem
- Innovation and uniqueness of the solution

DETAILED EXPLANATION — six domains on one platform

Flood & Fire

SAR sees through monsoon cloud; thermal SWIR pinpoints hotspots and sends the exact GPS grid.

Heat & Air

Land Surface Temperature drives street-level heatwave warnings, rain nowcast and AQI.

Crop Health

Weekly NDVI vigour score, SAR soil moisture for irrigation timing, early pest-stress signals.

Land Change

Yesterday-vs-today pixel differencing exposes illegal construction and lake encroachment.

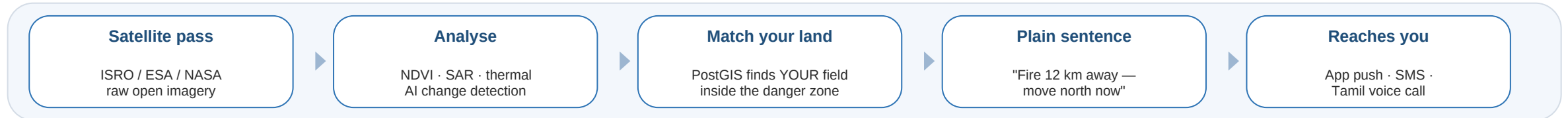
Sea & Fishing

Fishing zones from sea temperature and chlorophyll; offline alarm before the maritime border.

Environment

Oil spills, industrial effluent and deforestation, auto-escalated to the right authority.

HOW IT ADDRESSES THE PROBLEM — how one alert is born



INNOVATION AND UNIQUENESS

■ 0

Zero data cost

We launch no satellite. Public feeds are consumed through open APIs, so the core map stays free for the citizen forever.

100%

Reaches 2G phones

Every rival needs a smartphone and 4G. Our IVR and SMS path reaches the rural majority they structurally cannot serve.

6 → 1

One window, not ten

Ten disconnected agency portals collapse into one dashboard answering with a single plain Tamil sentence.

Working prototype today: live 3D Earth tracking 7,906 real satellites from live TLEs — verified to 0.3 km against independent ISS telemetry — plus full bilingual onboarding and the six-pillar dashboard.



TECHNICAL APPROACH

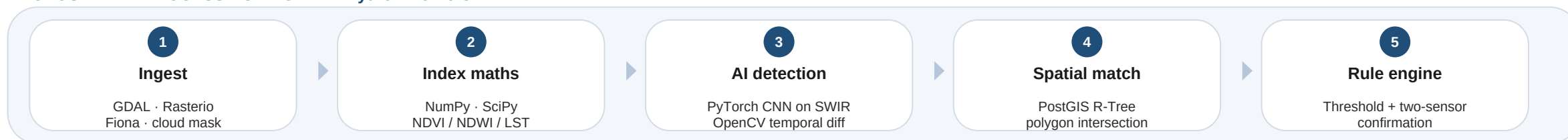


- Technologies to be used (e.g. programming languages, frameworks, hardware)
- Methodology and process for implementation (Flow Charts/Images/ working prototype)

1 · OPEN SATELLITE DATA SOURCES



2 · GEOSPATIAL PROCESSING ENGINE · Python workers



3 · DUAL DATABASE — each does what the other cannot

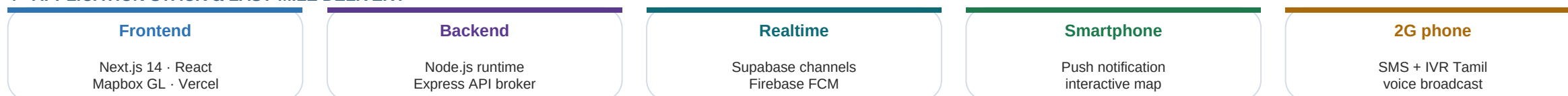
SUPABASE · the brain

PostgreSQL + PostGIS — an R-Tree spatial index answers "is this farmer's polygon inside today's flood mask?" in under a millisecond. Holds land boundaries, sea zones, alert history, auth and analytics.

FIREBASE · the mouth

FCM push + offline cache — Supabase Realtime only reaches an OPEN app. Only FCM wakes a CLOSED app at 2 a.m. when fire starts. Also carries offline sync at sea, where there is no signal at all.

4 · APPLICATION STACK & LAST-MILE DELIVERY



All-weather resilience: when monsoon cloud blinds optical sensors the pipeline auto-switches to Sentinel-1 C-band SAR, fused with INSAT-3D thermal refreshing every 15 minutes over India — so an alert never fails for want of a clear sky.



FEASIBILITY AND VIABILITY

- Analysis of the feasibility of the idea
- Potential challenges and risks
- Strategies for overcoming these challenges

ANALYSIS OF FEASIBILITY · can it be built today?



Technology is already in orbit

We build and launch nothing. Multi-billion-dollar public satellites are consumed through open REST APIs that are live right now.



Skills are mainstream

Next.js + Express + Node is a standard stack, and the Python geospatial libraries (GDAL, Rasterio, PyTorch) are mature and documented.



Zero capital expenditure

Only open-source packages and free tiers — Vercel plus Render carry the pilot. No licence fees and no hardware to purchase.



Legally clear

Fully within the National Geospatial Policy 2021: we parse only civilian-grade public telemetry released for public good.

CHALLENGES / RISKS → STRATEGIES TO OVERCOME

- | | |
|--------------------------------|---|
| • Cloud blocks optical imagery | → auto-fallback to SAR radar |
| • Satellite revisit time gaps | → fuse geostationary INSAT-3D |
| • Villages with no internet | → SMS + IVR voice pathway |
| • False alarms erode trust | → two-sensor confirm before dispatch |
| • Scaling compute cost | → tile caching + district batching |

VIABILITY · will it sustain long-term?

The need repeats every single year

Monsoons, crop cycles and fishing seasons return annually, so retention is structural — not a one-time utility people uninstall.

Free core, paid precision

Map and public alerts stay free forever. ■15–20/month buys custom-drawn asset boundaries and daily automated monitoring.

Government contracts (B2G)

State Disaster cells and municipalities license sprawl timelines and encroachment evidence on multi-year institutional terms.

A moat rivals cannot cross

Competitors are smartphone-and-4G only. Our IVR and SMS layer reaches the rural majority they cannot serve at any price.

SCALE PATH

- | | |
|----------------|--|
| Phase 1 | Tamil Nadu pilot — 2 districts, 5,000 users |
| Phase 2 | All coastal states — Tamil + 4 languages |
| Phase 3 | Pan-India — 12 languages, state DM integration |
| Phase 4 | Open API for panchayats, NGOs, researchers |
| Phase 5 | Institutional B2G licensing at state scale |

■0 CapEx to launch

7,906 satellites live now

0.3 km verified accuracy

2G minimum device

100% open data, no licences

- Potential impact on the target audience
- Benefits of the solution (social, economic, environmental, etc.)

POTENTIAL IMPACT ON THE TARGET AUDIENCE

Farmers

2.5 crore+ farm households

A weekly crop health score and irrigation timing replace guesswork. Land is walked once with GPS and saved forever. Pest stress is flagged before it spreads field to field.

Fishermen

40 lakh+ coastal livelihoods

Fishing-zone bearings before departure end wasted fuel on empty trips, and the offline maritime-border alarm prevents arrests and boat seizures far out at sea.

Families & workers

Every citizen in the area

Heatwave, air-quality and flood warnings for their own street — and for their hometown village if they work in another city, so family is never left unwatched.

Authorities

State DM cells, municipalities

Fire coordinates arrive automatically instead of by phone call. Encroachment gets timestamped satellite evidence. Urban sprawl is tracked year over year.

BENEFITS OF THE SOLUTION (social, economic, environmental)

SOCIAL

Lives protected

Warnings reach the poorest and least connected first instead of last, and language is never the barrier — the alert speaks Tamil, by voice if needed.

ECONOMIC

Losses prevented

Less crop lost to undetected pest and missed irrigation. Less diesel burnt on empty fishing trips. Fewer boats seized and fewer livelihoods wiped out.

ENVIRONMENTAL

Damage caught early

Faster fire response means smaller burns. Encroachment, effluent discharge and deforestation become visible and accountable, so they deter.

MEASURABLE OUTCOMES

< 10 s

from satellite detection to citizen alert

15 min

fire hotspot refresh cycle over India

100%

of the population reachable, 2G included

0

cost to the citizen, permanently

The real outcome: a farmer who cannot read a raster and a fisherman with no signal both receive the same intelligence a government analyst gets — as one sentence, in Tamil, on the phone already in their pocket.

- Details / Links of the reference and research work

OPEN DATA SOURCES WE BUILD ON

ISRO Bhuvan bhuvan.nrsc.gov.in	Cartosat, RISAT SAR, Oceansat, INSAT-3D/3DR thermal
ESA Copernicus dataspace.copernicus.eu	Sentinel-1 C-band SAR, Sentinel-2 10 m, 5-day revisit
NASA Earthdata earthdata.nasa.gov	Landsat 8/9 archive, MODIS, VIIRS active fire
INCOIS incois.gov.in	Potential Fishing Zone advisories, ocean state forecast
CelesTrak celestrak.org	Live TLE orbital elements powering our 3D tracker
Bhashini bhashini.gov.in	Government NLP stack for Indian-language alerts
BigEarthNet bigearth.net	Sentinel-2 benchmark corpus for land-cover training

VALIDATED METHODS & LITERATURE

NDVI	Rouse et al., 1974 — $(NIR-Red)/(NIR+Red)$ vegetation vigour
NDWI	McFeeters, 1996 — surface water delineation for flood staging
SGP4 / SDP4	Hoots & Roehrich, 1980 — NORAD orbital propagation
SAR change detection	Pixel-wise backscatter differencing for encroachment
PFZ methodology	INCOIS SST + chlorophyll convergence advisory model

EXISTING SOLUTIONS STUDIED · and how VaanEye differs

BigEarth.net / BigEarthNet EU benchmark dataset for Sentinel-2 land-cover classification. → A research corpus for scientists. We train on it, then turn the same imagery into a citizen sentence.
Google Earth Engine Planetary-scale raster computation inside a code editor. → Requires JavaScript and remote-sensing skill. VaanEye requires only a phone number.
Bhuvan / MOSDAC portals Official ISRO map viewers and data download portals. → Single-domain viewers, no personal alerting. We fuse six domains and push to the user.
Sentinel Hub / EO Browser Commercial imagery browsing and processing APIs. → Paid tiers aimed at analysts. We stay free at the point of use for the citizen.
mKisan / Meghdoot Government SMS agro-advisory services. → Generic advisory text with no live satellite verification of the user's own field.

SIH 2026 SPACE-TECH CONTEXT & PROTOTYPE STATUS

✓ Related SIH 2026 PS studied SIH26227 multi-temporal change analysis · SIH26192 flash-flood prediction · SIH26191 hazard red zones
✓ Live 3D Earth prototype 7,906 real satellites, SGP4 in-browser, verified 0.3 km vs live ISS telemetry
✓ Bilingual onboarding built 6 steps Tamil + English, GPS land walking, sea-zone picker, multi-location watch